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How autonomous research shapes scientific understanding

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Temporary ImageGen figure preview. Generated geometry is illustrative; quantitative evidence is in the text and tables.

Autonomous science promises to translate experimental agency into advances in scientific understanding[1,2]. Whether and how self-directed experimentation realizes this promise requires examining the research process itself. Here we use ChemWorld, a set of programmable chemical worlds, to connect experimental choices, observations, contemporaneous judgments and subsequent predictions. Broad comparisons across chemical systems motivate repeated equilibrium and crystallization studies, with 90 current process sessions and a separate 30-session comparison of research opportunity. Initial information is associated with distinct experimental starting points; ensuing observations support prospective tests, local relationships and explicit revisions. Yet the consequences of research depend on how these relationships are subsequently used. In the matched research-opportunity comparison, access to research lowers average prediction error by 73% overall while increasing it by 69% at extreme dilution. Crystallization reveals a complementary divergence: forecasts improve on a source-observation mean for recovery in 39 of 42 sessions, but in none for purity. Continuous records locate transitions between supported local judgments, further tests and extrapolations that remain unresolved by the acquired evidence. These findings show how scientific progress can differentiate within a research history. Experimental achievement, predictive generalization and evidential support describe distinct outcomes of autonomous inquiry.

An autonomous researcher does more than act on the world. By choosing experiments and measurements, it also shapes the experience from which its scientific judgments form: which phenomena become visible, which explanations encounter a test and which results guide the next investigation. The promise of autonomous science therefore concerns both the expansion of experimental capability and the transformation of experimental experience into scientific understanding.

Robot scientists and self-driving laboratories have demonstrated hypothesis testing, materials discovery and reaction optimization[3–6]. Language-model agents extend the connection between scientific instructions, tools and laboratory operations[7,8]. These advances make it possible to investigate a further question: how does an agent's own research change the relationships it uses to explain and predict the world? A successful procedure establishes an experimental achievement, and a correct forecast establishes predictive accuracy. Neither alone reveals the evidence from which a judgment arose or the scope that evidence supports[9,10].

We study this process in ChemWorld. Programmable world laws and independently specified public information make it possible to examine how initial knowledge enters experimental choices, how observations support or change judgments, and how these judgments guide further inquiry. Our evidence follows the organization of research through the formation of working relationships to their use under new interventions. Successful local tests, warranted revisions, unsupported interpretations of execution and response-specific transfer need not share a single failure mechanism. Together they reveal how the scientific consequences of research depend on the transitions between choosing evidence, interpreting it and applying what it supports.

Making autonomous scientific inquiry experimentally tractable

ChemWorld combines materials, operations and instruments into stateful virtual chemical systems. Agents choose quantities, processing sequences, measurements, repetitions and subsequent experiments within a fixed resource allowance. Their choices change samples and determine which observations become available. An evaluator independently controls world parameters and public information, records committed operations and obtains withheld reference outcomes (Fig. 1). The agent does not receive those outcomes during research or assessment.

This arrangement makes the agent's research process experimentally accessible. Knowledge tests and supplied-data tasks assess important scientific abilities[11,12], and interactive discovery environments already connect action to observation[13–16]. ChemWorld connects an agent's evidence selection to evolving public judgments and their later use in composable chemical processes. An attempted operation can be rejected before changing the sample; an observation can support a relationship locally without establishing it under every compatible intervention. Execution records and independently evaluated world responses allow these distinctions to be checked.

The broad study comprises 240 Sol campaigns across six chemical families and 60 additional equilibrium campaigns from four other model configurations. The current process analysis combines 45 equilibrium sessions with 45 crystallization sessions conducted under a corrected, consistent recovery definition. Each system covers five fixed worlds, three information conditions and three fresh model sessions per condition. These 90 sessions contain 80 complete research-and-assessment chains and 83 valid forecast stages; incomplete sessions remain in the denominator. Earlier crystallization sessions affected by

a recovery-description mismatch are retained separately, rather than pooled with their replacement cohort.

Contemporaneous notes connect publicly stated judgments to the observations available at that time. After research, agents seal a scientific account, predict outcomes under twelve new interventions without laboratory access, and reflect without receiving reference truth. Structured process analysis identifies 92 observable judgment–experiment–revision windows across 65 of the current 90 sessions (Extended Data Fig. 1). The other 25 sessions lack a complete window under this recording definition; this does not establish an absence of scientific learning. A separate contemporaneous comparison assigns 30 fresh sessions to research or no research under a common assessment protocol (Fig. 4). These evidence layers answer complementary questions and retain separate denominators.

The self-organization and adaptation of scientific inquiry

The organization of inquiry begins with a goal, available information and a constrained set of actions. These conditions shape the research history, but do not determine it in a simple way (Fig. 2). Across 60 historical electrochemical campaigns, discovery and optimization instructions do not produce a consistent division between broad exploration and narrow local search. With twelve batches, mean solvent coverage is 3.93 and 3.73, respectively, and intermediate measurements average 12.00 and 10.53. With twenty-four batches, both goals cover all four solvents and use all twenty-four intermediate measurements. These coarse descriptions do not explain the full difference between the resulting procedures or predictions.

Initial information has a clearer relationship to the starting point of the corrected crystallization studies. All fifteen aligned-information sessions begin with solvent S2 and catalyst C1; all fifteen sessions without an additional instance dossier begin with S0 and C0. Under the specified mismatched dossier, thirteen of fifteen begin with S2 and fourteen with C1; one session ends before a catalyst is successfully added. Mean solvent coverage over the research history, including committed resuspension additions, is 4.00 without the dossier, 2.93 with aligned information and 2.40 with mismatched information. These are concrete differences in the evidence agents choose to acquire. Broader coverage is not itself evidence of better understanding, and the comparisons do not isolate coverage as a cause of subsequent error.

Observations can then change the direction of research. In one equilibrium session from the research-opportunity comparison, the initial note predicts a weak-acid trend: increasing concentration should lower pH and dissociation fraction. The first six experiments instead yield dissociation fractions near 0.064–0.079. The next note withdraws support for the expected trend over that range and proposes that another accessible component might be the relevant input. Experiments eight to twelve actually use that component, yet still yield a near-plateau response. The final account retains a bounded empirical relationship while leaving the hidden mechanism unresolved.

This sequence links an initial expectation, contrary observations, a proposed alternative and a changed experiment. Other studies continue a scan already planned at the outset, narrow a parameter interval or repeat a borderline recipe. All are forms of organizing inquiry, but only some constitute newly proposed tests in response to feedback. In the corrected crystallization cohort, 52 of 58 selected immediate plans match the ensuing committed actions and six match partly. This measures a public plan–action connection within observable windows; it is neither a discovery rate nor evidence that writing the note caused the action. The next question is what those actions and observations establish.

The formation and evolution of scientific understanding through experimentation

Scientific relationships acquire practical meaning when they support a further test, constrain an explanation or guide a new intervention (Fig. 3). In a corrected crystallization study, a near-threshold fines result leads to a prospective proposal to lower the reagent charge from 0.03 to 0.02 mol. The next batch implements that proposal and passes the fines threshold. The result supports a local recipe improvement. Because the absolute catalyst amount remains fixed, however, the catalyst-to-reagent ratio also changes: the experiment does not separately identify a concentration mechanism.

A second study makes an interpolation. After observing different recovery and fines outcomes at nearby solvent volumes, the researcher proposes that 0.065 L should preserve the recovery benefit while improving fines. The proposed experiment fails, and the final note retracts the expectation. Here, an unsuccessful operating intervention produces an observable revision of a scientific judgment. A further near-threshold study explicitly discusses measurement noise, repeats a 310 K procedure with two waiting stages, and still recommends a recipe that fails retesting. Recognition of uncertainty, purposeful repetition and operational failure can coexist.

Whether an intervention actually occurred is itself part of its evidential status. One researcher plans approximately 1 g of cumulative seed addition, but only 0.05 g is committed and nineteen subsequent additions are rejected. Its later note correctly states that the attempted sequence cannot establish the effect of the high seed dose. Another session encounters analogous rejected additions yet describes 1 g as the tested condition and uses the outcome against a seed-based explanation. The contrast concerns the transformation of an execution history into scientific evidence: the world did not perform the intervention that the latter account treats as a negative test.

A related study attributes low recovery after heating to 440 K to a thermal-history boundary and subsequently tests 430 K. The later test is real, but the purported earlier comparison includes cooling requests to 265 and 270 K that were rejected. The sequence contains adaptation and further experimentation without establishing the thermal contrast claimed in the explanation. Action, observation

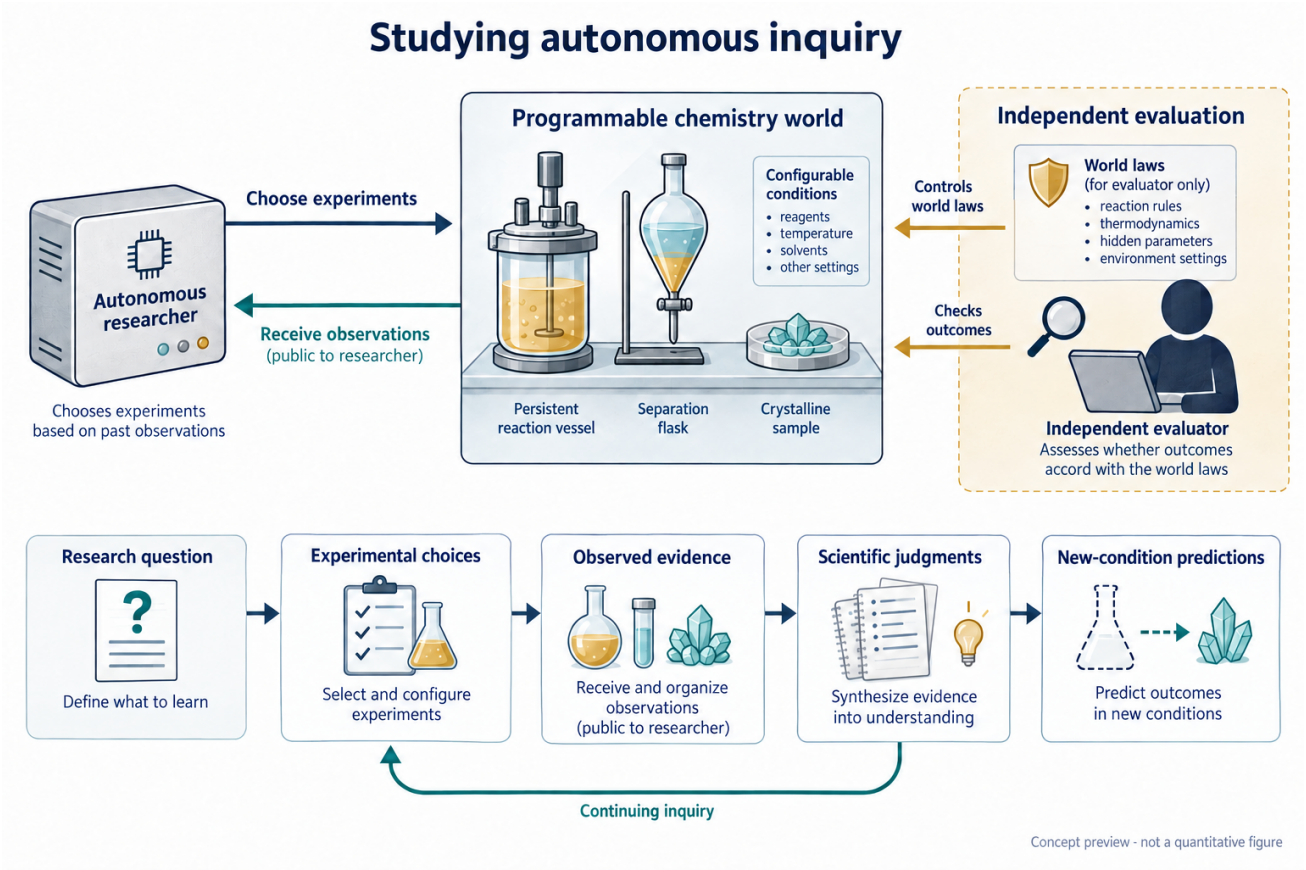


Figure 1 | Making inquiry experimentally tractable. Temporary ImageGen concept illustration. An autonomous researcher selects interventions and receives public observations from a persistent chemical world. The evaluator separately controls hidden parameters and checks world outcomes; reference outcomes are not returned to the researcher. The lower sequence links questions, choices, observations, public judgments and predictions, with judgments informing further choices. It summarizes observable relations, not a direct measurement of private cognition or a demonstrated causal effect of notebook text.

and interpretation must therefore be connected individually rather than inferred from a coherent final report.

The equilibrium study introduced above makes a different transition. Its component tests appropriately weaken one explanation of the local plateau. When forecasting an extremely dilute condition, however, the agent writes: "The point estimate still favors the experimentally supported fixed acidic background". It predicts dissociation of 0.085 with an 80% interval of 0.035–0.350; the withheld reference mean is 0.716. Both the scientific account and forecast recognize that the distant regime was not tested. The issue is not simply a failure to mention limits. A locally supported relationship receives substantial weight in a prediction whose continuation the experiments have not resolved.

That continuation is not inevitable. Two sessions in another fixed equilibrium world both observe plateaux at relatively high loading, yet their dilute macro errors are 0.16287 and 0.01462. For the same dilute query, they predict dissociation of 0.130 and 0.626 against a reference mean of 0.65463. The more accurate session uses a weak-acid continuation consistent with its plateau observations and explicitly describes it as conjectural. Accurate extrapolation does not require direct sampling of every target condition; neither does accuracy establish that research experimentally identified the successful explanation. Continuous records distinguish supported local relations, unresolved possibilities and their subsequent predictive use.

The nature and conditions of scientific progress in autonomous inquiry

The process records locate changes in judgments; the separate research-opportunity comparison tests what the opportunity to investigate contributes to subsequent prediction (Fig. 4). Five equilibrium worlds are each studied by three fresh sessions with twelve experimental batches and three without source experiments. Both groups receive the same scientific public context and the same sealed assessment. All thirty sessions terminate, all 180 planned source batches complete, and twenty-nine sessions provide valid forecasts. One research session loses its forecast and later stages to a network interruption; its source observations and scientific account remain retained.

Research substantially improves prediction overall. Averaging valid sessions within each world and then weighting the five worlds equally, normalized macro mean absolute error falls from 0.132086 without research to 0.035632 with research, a 73.0% reduction. The direction holds in all five worlds. Yet for the three prospectively designated extreme-dilution queries, error rises from 0.061467 to 0.103677, a 68.7% increase, with deterioration in four of five worlds. Over the remaining nine queries it falls from 0.155625 to

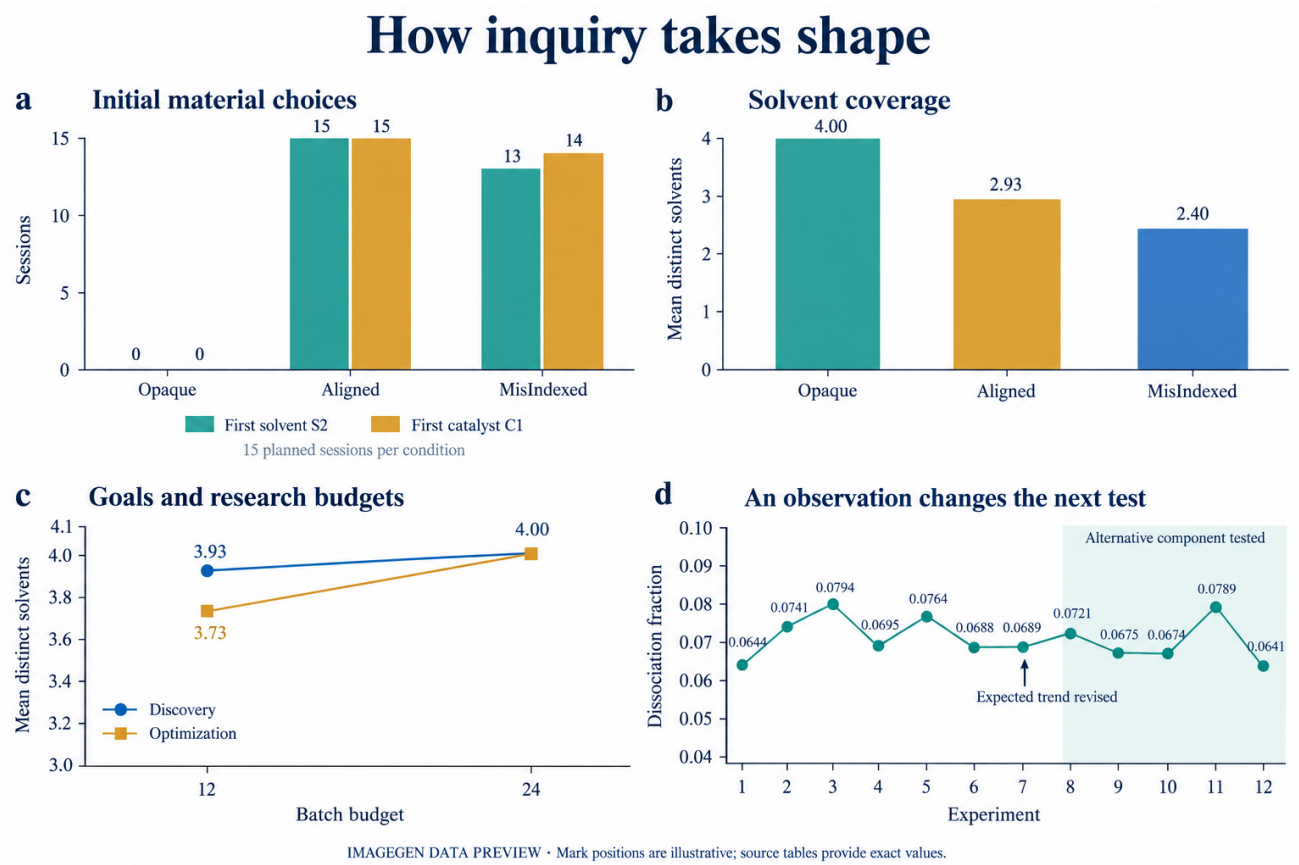


Figure 2 | How inquiry takes shape. ImageGen data preview. a, First use of solvent S2 and catalyst C1 among fifteen corrected-crystallization sessions per information condition: Opaque 0/0, Aligned 15/15 and MisIndexed 13/14. b, Mean solvent coverage is 4.00, 2.93 and 2.40, respectively. c, Historical electrochemical discovery and optimization cover 3.93/3.73 solvents with twelve batches and 4.00/4.00 with twenty-four. d, A selected equilibrium trajectory records dissociation fractions of 0.064–0.079, revises its initial weak-acid expectation after the first six batches, and tests an alternative component during batches 8–12. These are descriptive choices and a recorded sequence, not an isolated causal effect of information or notes. Generated geometry is illustrative; exact observations are retained in the source records.

0.012950. The comparison concerns the total opportunity to conduct research, including experiment selection, feedback and available recording tools; it does not isolate any one component.

The same opportunity can therefore improve average prediction while worsening transfer in a defined region. The continuous equilibrium case connects a revision induced by observations to the later weighting of a local relationship in a distant forecast. Its no-research counterpart predicts dissociation of 0.569 for the query above, closer to the reference, but explicitly relies on an untested prior assumption. That better forecast is not itself an experimentally established discovery.

The result extends beyond point error without yielding a universal story of misplaced confidence. At extreme dilution, research produces lower empirical coverage than no research for all three nominal 80% intervals, while the no-research intervals are substantially wider. An interval score penalizing both width and misses favours research for pH, favours no research for dissociation fraction and is similar for precipitation signal. Across all queries, research improves interval scores for all three responses. A conservative sensitivity analysis allowing the missing session's macro error to take any value in its legal range does not reverse either the overall benefit or the dilute deterioration (Methods).

Crystallization exposes the uneven use of evidence already acquired (Extended Data Fig. 2). For each of the 42 valid forecasts, we compare the agent with transparent references using only that same session's source observations. Five-world-equal purity error is 0.03853 for the agent and 0.00510 for the overall source mean; the source mean is more accurate in every valid session. Recovery shows the opposite pattern: agent error is 0.10129 against 0.20443, with the agent outperforming the mean in 39 of 42 sessions. Conditioning the reference on the exact committed solvent set preserves both aggregate directions. Size and fines differ again, so the response channels are reported separately rather than collapsed into a common learning score.

These observations complement earlier operating–predictive comparisons. In electrochemistry, optimization produces a better recommendation in 26 of 30 matched conditions; among those 26, prediction is better in thirteen and worse in thirteen. In the repeated equilibrium cohort, aligned partial knowledge reduces error over nine queries from 0.00907 to 0.00512 while increasing dilute error from 0.09880 to 0.13205. Initial information, experimental

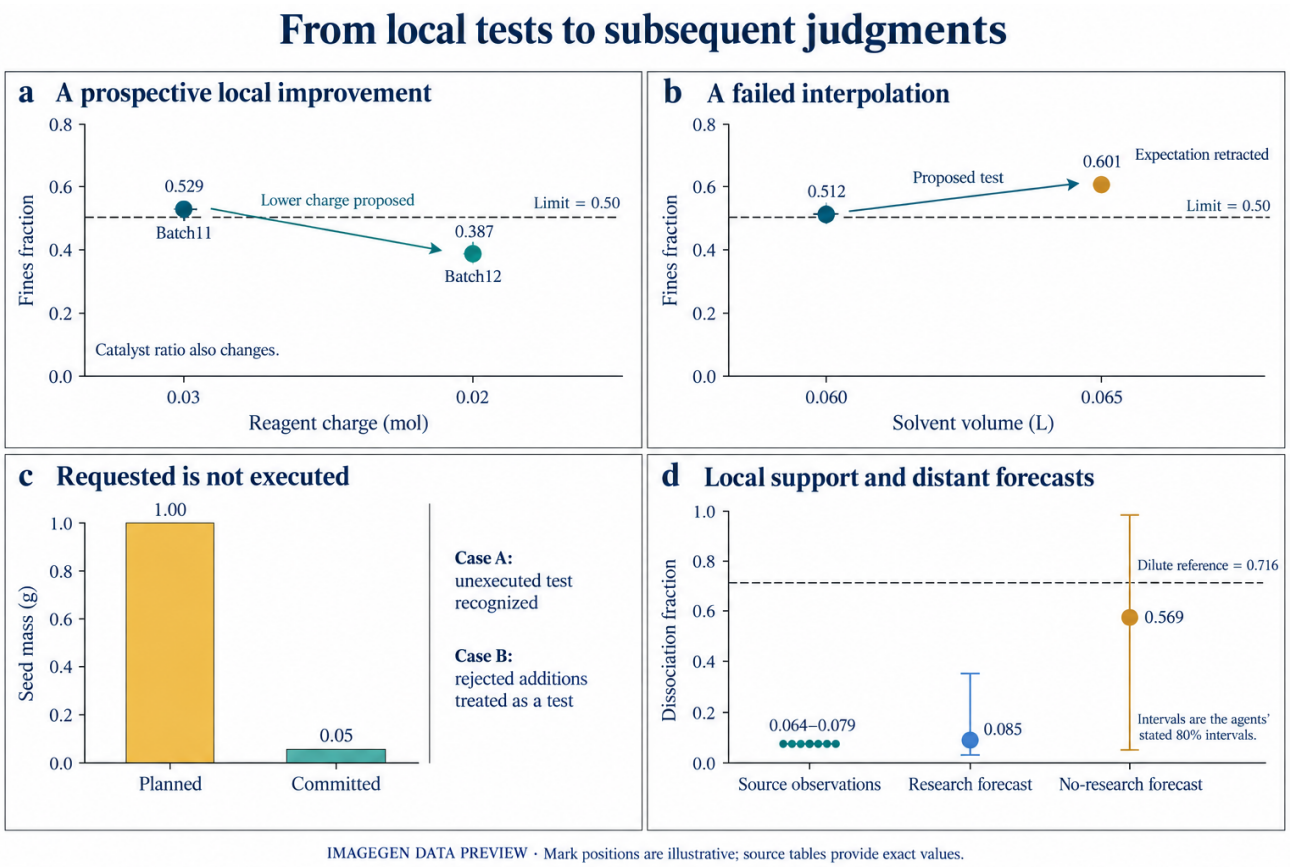


Figure 3 | From local tests to subsequent judgments. ImageGen data preview of selected cases. a, Reducing reagent charge from 0.03 to 0.02 mol lowers fines from 0.529 to 0.387, crossing the 0.50 threshold; the fixed catalyst charge also changes the catalyst ratio. b, A proposed solvent-volume interpolation from 0.060 to 0.065 L instead increases fines from 0.512 to 0.601 and is retracted. c, Requests for 1 g seed leave only 0.05 g committed; separate researchers recognize the unexecuted test or describe it as negative evidence. d, The equilibrium source plateau is contrasted with a dilute-condition forecast: Research 0.085, interval 0.035–0.350; matched NoResearch 0.569, interval 0.040–0.980; withheld reference 0.716. Intervals are the agents' stated 80% predictive intervals. Source dots and chart geometry are illustrative, not an exact empirical point inventory.

achievement and subsequent prediction each contribute to the scientific outcome, but none substitutes for examining how research supports a relationship and how that relationship is used.

Discussion

Autonomous inquiry organizes both interventions in the world and the evidence on which subsequent judgments depend. ChemWorld makes the connections between these activities observable: initial information enters choices; observations support local relations and revisions; those relations guide further tests and predictions. The principal finding is that progress can differentiate within this process. An experiment can establish a useful local relationship without resolving its continuation elsewhere, and one research history can support markedly different predictive outcomes across response variables.

Studying the formation of scientific understanding requires specifying which relationship is proposed, what information supports it at that point, what the world actually tests and how the relationship is later used. The records include warranted retractions and bounded improvements alongside interpretations unsupported by committed operations. They also include accurate forecasts grounded in possibilities that remain experimentally unresolved. These distinctions are consistent with the role of discriminating tests and inference in scientific inquiry[17–19], but their implementation by autonomous agents cannot be assumed from successful tool use or a plausible scientific account.

The controls delimit what can be explained causally. The matched opportunity comparison estimates the effect of the full research package under a common assessment protocol. A separate fixed-recipe quench intervention identifies physical effects of an operation in the virtual worlds, without providing outcomes to an adapting agent (Extended Data Fig. 3). Neither identifies a notebook's independent effect or proves that a particular judgment maintains itself through selective experimentation. Explaining the observed formation and use of relationships does not require every trajectory to share such a mechanism or end in a successful corrective strategy.

The study is limited to selected programmable systems, a small fixed set of worlds and, for repeated process analyses, one model configuration. Fresh sessions are not controlled model-seed replications or independent samples of world laws. Notebook statements are public records rather than access to private cognition; gaps restrict temporal inference. Cases are retrospectively selected from system-

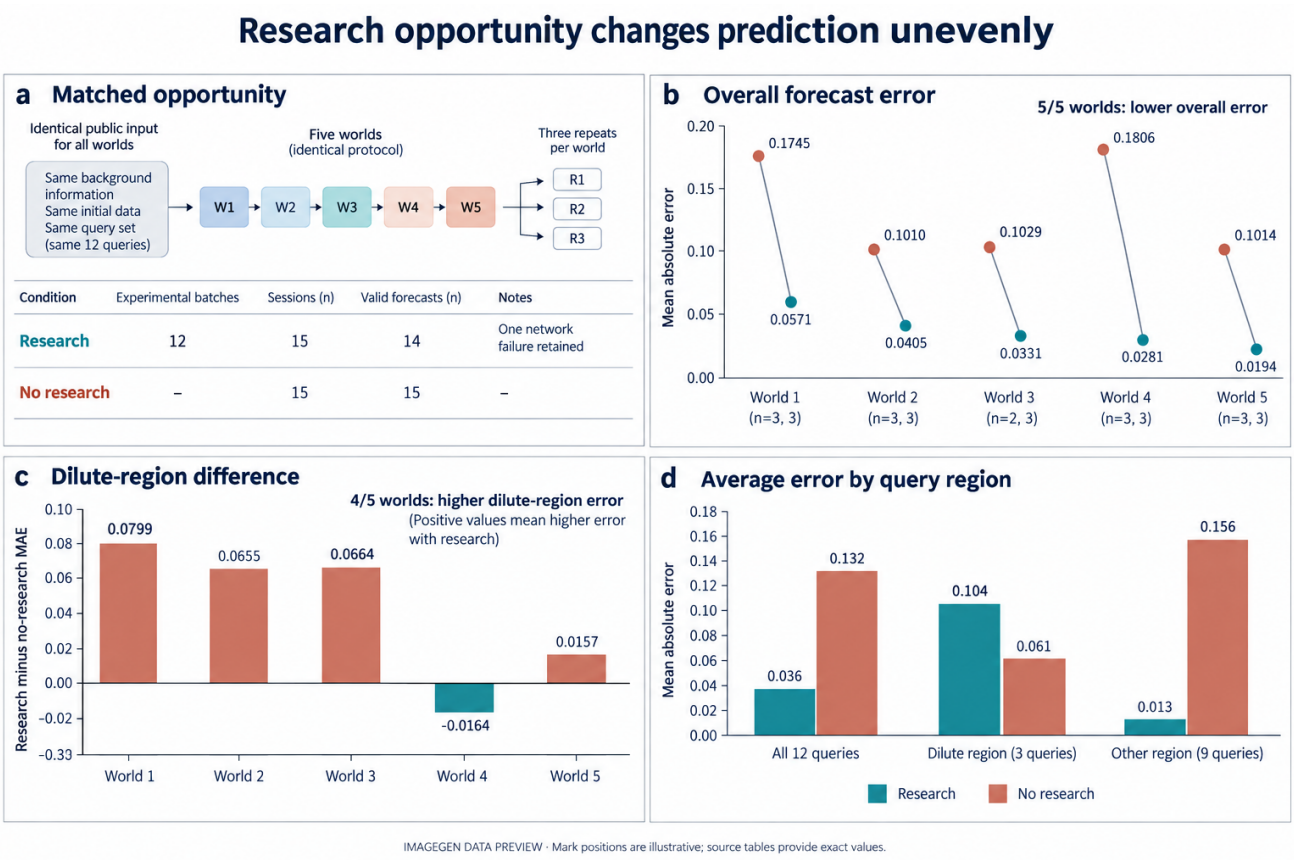


Figure 4 | Research opportunity changes prediction unevenly. ImageGen data preview. a, Common public inputs and assessment are matched within fifteen Research/NoResearch pairs across five worlds; Research allows twelve source batches. Fourteen Research and fifteen NoResearch forecasts are valid; one network failure is retained. b, Overall world-level mean absolute error is lower with research in all five worlds. c, The Research-minus-NoResearch difference on the three dilute queries is positive in four worlds. d, Five-world-equal errors for Research/NoResearch are 0.035632/0.132086 across all twelve queries, 0.103677/0.061467 for the dilute subset and 0.012950/0.155625 for the other nine. World 3 has two valid Research forecasts; other cells have three. Generated marks are illustrative; the comparison estimates access to the full research package.

atically retained records, with single-analyst interpretation and no second-rater reliability estimate. Historical and current protocols retain separate provenance, including the corrected crystallization block and all failures. These limits constrain claims about prevalence and generality while leaving the observed transitions and bounded comparisons available for scrutiny.

The scientific promise of autonomy concerns how self-organized investigation changes the relationships that guide understanding of the world. By connecting choices, evidence, judgments and subsequent use, ChemWorld turns this transformation into a tractable object of study—one in which experimental progress, warranted understanding and reliable transfer can be examined together without being treated as the same achievement.

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Methods

Worlds, interfaces and experimental units

ChemWorld compiles compatible process components into persistent virtual chemical environments. The public contract specifies materials, legal operations, instruments, resources and task objectives. Private parameters determine each world's responses. Agents access the laboratory and public records, without evaluator state, simulator code, private parameters or withheld reference outcomes. Operations change samples; measurements may consume material. Candidate transitions are validated before commitment. Rejected operations do not constitute the requested physical intervention, although attempts can consume contract-defined allowances. Exact replay checks environment transitions, not regeneration of model decisions.

The broad study covers electrochemistry, partitioning, reaction and thermal processing, aqueous equilibrium, reaction-coupled crystallization and purification. These are controlled virtual systems, not experimentally validated predictions for named physical substances. Current repeated studies use equilibrium and crystallization. Five fixed worlds recur across arms and repetitions; their parameters, initial states and keyed observation mappings are fixed. Repetitions are fresh model sessions, not independent world draws or controlled model-seed replications. The model interface does not expose a usable sampling seed. Repeated identical interventions can share keyed observations, so they are not independent noise samples.

Evidence layers and retained denominators

The historical broad pool contains 240 Sol-medium campaigns and 60 additional equilibrium campaigns from Luna, Terra, GPT-5.5 and Astra, requested at medium reasoning effort. The five-model equilibrium comparison includes fifteen Sol campaigns already in the 240. Requested model labels do not establish equal realized computation. The original 240 include 60 electrochemical, 30 partitioning, 60 reaction, 45 equilibrium, 30 crystallization and 15 purification campaigns. They retain 3,597/3,600 final assays, all 720 assessment stages and 238 protocol-conforming campaigns under their original rules. The additional sixty retain 720 batches and 240 stages. Two original source-completion failures remain included. Earlier selected notebook studies comprise another 24 sessions and are not pooled into the broad or current process cohort.

The current process cohort combines the 45 equilibrium sessions from the earlier 48-session block and its 42-session extension with a complete replacement block of 45 crystallization sessions under a consistent recovery description. Each system covers five worlds, Opaque/Aligned/MisIndexed information and three sessions per cell, with twelve research batches per session. Equilibrium retains 39 complete chains and 41 valid forecasts; corrected crystallization retains 41 and 42, respectively. The combined current denominator is 90 sessions, 80 complete chains and 83 valid forecasts. Old crystallization sessions remain protocol-affected historical evidence. They are not added to the replacement sessions when calculating current outcomes or process frequencies.

Corrected crystallization has 510/540 source batches, 125/135 assessment stages, 145 notebook versions and 42 recommendation retests. All 45 source trajectories and 42 retests pass replay. Four incomplete chains remain: three lack forecasts, and one retains a valid forecast but lacks the subsequent retrospective. The equilibrium subset and the separate opportunity block retain their own failure tables. Missing outcomes are not set to zero, and later favourable outcomes do not replace completed results. Evidence integration does not change the original development or qualification status of any block.

Information, autonomy and recording

Opaque supplies the public task and operational context without the additional instance dossier. Aligned supplies partial information appropriate to the system. In equilibrium, this includes an effective-acidity interval and an anchored local dilution relationship. MisIndexed uses another world's archival relationship with the same format and qualification. In crystallization, Aligned gives nominal descriptors for anonymous materials; MisIndexed swaps solvent S1/S3 descriptors while preserving S0, S2 and catalyst descriptors. Neither dossier specifies the complete world law.

The original equilibrium Opaque/Aligned sessions precede its MisIndexed extension. Corrected crystallization is a later complete block. These descriptive comparisons are not all concurrently randomized treatments, and new versus old crystallization is not an estimate of the causal effect of correcting the prompt. The new opportunity comparison uses a common contemporaneous protocol, described below.

Agents choose operations, quantities, measurements, repetitions and grouping within fixed budgets. Equilibrium permits twelve source batches, 180 operation attempts, twelve intermediate instruments and twelve final assays; crystallization uses its fixed twelve-batch materials envelope and 720-attempt ceiling. A notebook role asks for an initial testable expectation when available; subsequent updates are agent-selected, without a write-every-batch requirement or automatic reinjection. The role may influence research and is not assumed passive. Notes retain version and public-event cursor. These public records do not expose private cognition. Absence of a notebook-read tool call does not establish absence of access to the conversation's earlier content.

Assessments and recovery definition

After source research and any operating recommendation, agents seal a scientific account, answer twelve fixed new-intervention queries without laboratory tools and then reflect without truth feedback. Equilibrium includes a structured supplemental assessment. The evidence notation K1/Q/K2/EQS denotes these stages. Queries are hidden during research; post-assessment proposals are not treated as executed source experiments.

Equilibrium errors compare predictions with the mean of five reference observations per query. Coverage checks all five observations. Crystallization uses the noiseless full-process response immediately before final-assay sampling; source observations are noisy. The corrected block retains sixty reference batches, numerically matching the earlier reference outcomes, and separately retests each available committed recommendation.

The corrected recovery definition is seed-excluded recovered solid target divided by the tracked initial reagent charge, in moles, clipped to [0,1]. Source instructions, prediction prompts, observations and scores now refer to this same full-process quantity. The internal crystallization-stage metric instead divides by target product before separation. Earlier process instructions described this latter denominator while observations and scoring used initial reagent charge. Original scores are preserved as protocol-affected diagnostics; they do not enter the corrected cohort. The new block was rerun in full rather than retrospectively relabelling those scores.

Quality requires particles, purity at least 0.80 and fines at most 0.50; joint recommendation success also requires corrected recovery at least 0.10. Corrected joint counts are 13/15 Opaque, 10/14 Aligned and 10/13 MisIndexed among available retests, against fifteen planned sessions in each arm. Particle size is the public index min(d50/250 micrometres, 1); its error is not measured in micrometres.

Research paths and process interpretation

Retrospective path extraction covers sixty historical electrochemical, forty-five equilibrium and forty-five corrected crystallization sessions. Fifty within-world condition contrasts retain goal, budget, information and cohort. Counts use committed operations; solvent coverage includes additions during resuspension. Rejected attempts do not establish successful material use. Interrupted sessions remain in the planned denominator.

A structural process window connects consecutive public judgments with intervening executed experiments and new observations. Systematic extraction identifies 34 equilibrium windows across thirty sessions and 58 corrected crystallization windows across thirty-five sessions. All 92 are coded; fifteen equilibrium and ten crystallization sessions lack a complete observable window. Windows are nested within sessions and are not independent participant replicates or a denominator for learning success.

Interpretation distinguishes prior expectations, newly proposed versus already planned tests, committed actions, observations available at the next statement, revisions, supported scope and subsequent use. The corrected crystallization analysis checks 116 exact quotations and classifies the selected immediate plan as matched in 52 windows and partly matched in six. This is an action-link measure, not a causal effect of notes. Selected focal cases also connect sealed forecasts to source records. Quote locations, operation cursors and case-specific scope limits are retained in the supplementary evidence organization.

The same analyst performs coding and internal scientific review. There is no independent second-rater agreement estimate, blinding or claim of adjudicated mechanism prevalence. The team has assigned review to this workflow; completed internal review is distinguished from independent replication. Counterexamples include warranted retractions, correct recognition of unexecuted tests and accurate but explicitly conjectural extrapolation. Selection of illustrative cases is retrospective.

Source-only forecast references

Two transparent references are computed for each valid corrected-crystallization forecast using only that researcher's successful source final assays. The first averages all available assays. The second averages assays whose exact set of committed solvent identifiers matches the query, falling back to the overall mean when none match. Solvent sets include successful solvent additions and resuspension, excluding rejected actions. The fitting function receives no test truth. These references are descriptive calculations, not separately trained agents or mechanistically complete baselines.

There are 42 valid sessions and 504 queries. Same-solvent fallback occurs in 186/504 queries: 1/180 Opaque, 95/168 Aligned and 90/156 MisIndexed. Matching does not control thermal history, quantities or final liquid composition. References have no predictive intervals. Each response is reported in its own public scale; no combined score is formed across recovery, purity, size and fines.

Matched research-opportunity comparison

The fixed design contains five equilibrium worlds, Research/NoResearch and three fresh Sol-medium sessions per cell, for thirty planned sessions. Both arms receive the same task, materials, legal operations, instruments and initial state. Fifteen paired checks confirm identical scientific public inputs after excluding only the random campaign identifier. Research

additionally receives twelve autonomous source batches and the existing notebook opportunity; NoResearch receives no source operations or observations. Both arms use identical assessment prompts, schemas, order and restrictions.

All thirty sessions reach terminal status. Research completes 180/180 batches; the combined arms complete 117/120 assessment stages and provide 29 valid forecasts. In world three's third Research repetition, the forecast stream disconnects after the scientific account. Q and the two later stages are missing, without replacement. Thus Research has fourteen valid forecasts, NoResearch fifteen; Research world three has two rather than three. Fifteen source trajectories and twenty-five reference trajectories (300 reference batches) pass exact replay. The retained resource ledger is complete for 29/30 sessions; one has only recorded partial usage. Recorded input plus output totals 13,428,656 tokens. Missing usage is not zero and monetary cost is unavailable.

A first execution attempt failed because the assessment transport configuration omitted a required field. Four launched sessions, two completed source histories totalling twenty-four batches and their resource costs are retained separately; no scientific-account stage completed. After fixing that execution defect, the whole fixed thirty-session block restarted. This was a protocol repair, not replacement based on scientific outcomes. Only the complete restarted block enters the comparison.

The comparison estimates the total research-opportunity package under a shared assessment. It does not isolate autonomy relative to supplied data, feedback alone, notebook use or private representational change. It is between fresh sessions, without a within-session pretest. World and arm order are balanced in a predetermined launch schedule, not a random sample of world laws.

Quantitative summaries and missingness

Equilibrium macro MAE equally weights normalized pH, dissociation fraction and precipitation signal across the designated queries. The extreme-dilution group comprises three queries fixed before the repeated blocks; the remaining nine are not assumed entirely in distribution. Session errors are averaged within each world, then the five worlds receive equal weight. Individual sessions and all missing stages remain visible. Responses, queries, reference observations and batches are correlated within sessions and worlds; they are not independent researcher samples.

Coverage and mean interval width accompany the proper interval score for the agents' nominal 80% intervals[20]. With lower endpoint l, upper endpoint u, outcome y and alpha = 0.2, the score is (u-l) plus (2/alpha)(l-y) when y is below l, or (2/alpha)(y-u) when y is above u. Lower is better. Aggregates weight worlds equally. Agent intervals are not confidence intervals for our group means.

A post hoc sensitivity analysis assigns the single missing opportunity forecast any macro MAE in [0,1], the legal error range, and recomputes means using the planned three sessions per arm/world and five equal worlds. Research-minus-NoResearch bounds are [-0.098659, -0.031992] overall and [+0.036768, +0.103434] for dilution. These are conservative bounds on missing-outcome influence, not imputation, confidence intervals or population inference. Three repetitions per fixed condition describe variation without establishing a population reliability rate.

Matched quench intervention

A separate retained block applies the eighteen original first-cohort crystallization recommendations in five worlds, with and without a single quench operation: ninety pairs, 180 trajectories. All other action arguments and initial states are matched. Quench is removed at its recorded location or inserted immediately before the first seed/crystallization-cooling operation. No agent adapts or receives these outcomes.

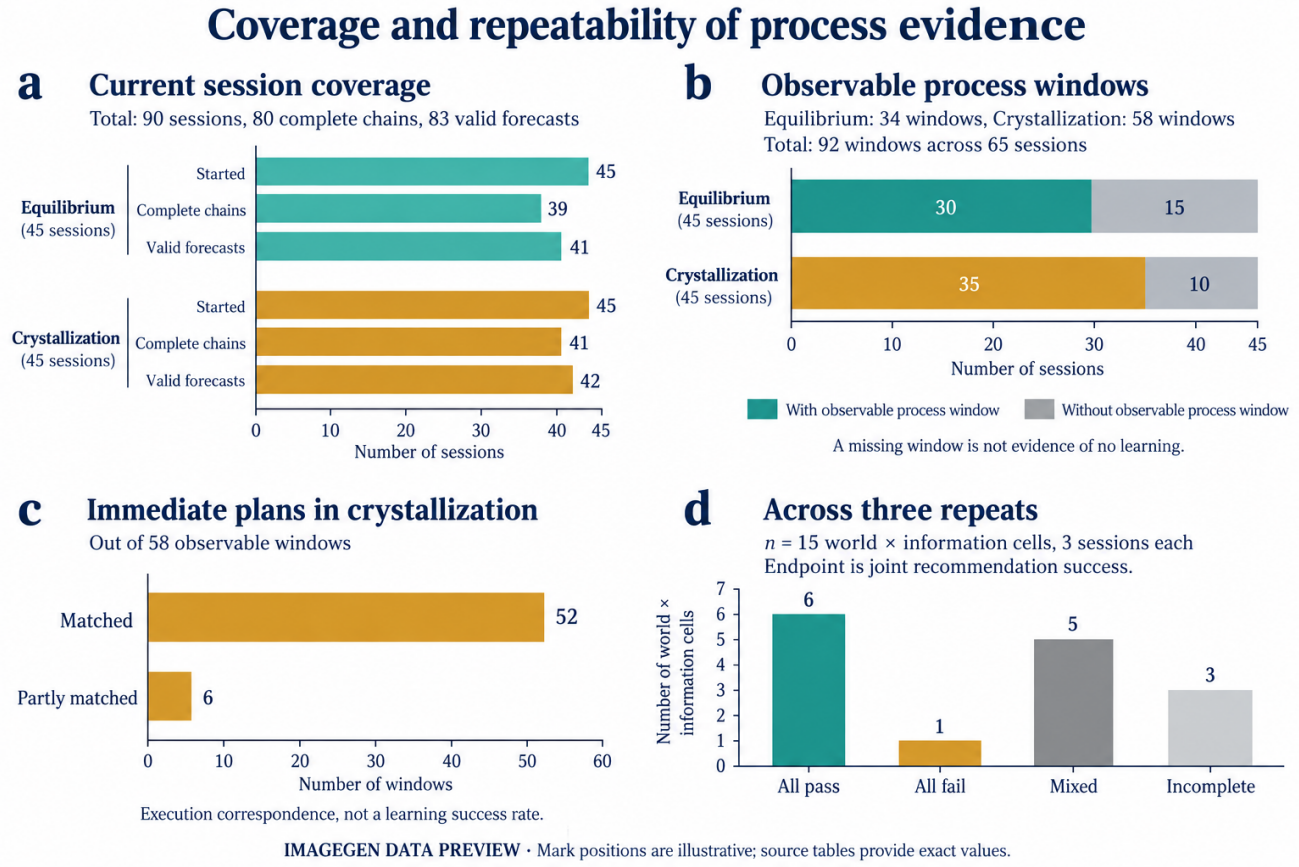
All trajectories are retained and replayed; seventy-five pairs reach both final assays. In five, the absent arm lacks particles; in ten, the present arm makes the subsequent fixed cooling request invalid. Failed arms are not repaired. Among complete pairs, mean present-minus-absent fines changes by +0.30937 and recovery by +0.00741; joint success changes from 34 to 4. These conditional effects describe physical operations under the implemented initial-charge recovery metric. Quench changes multiple physical state variables and does not isolate cooling rate or a cause of scientific judgment. Donor recipes are historical and are not sampled from the corrected block.

Data and code availability

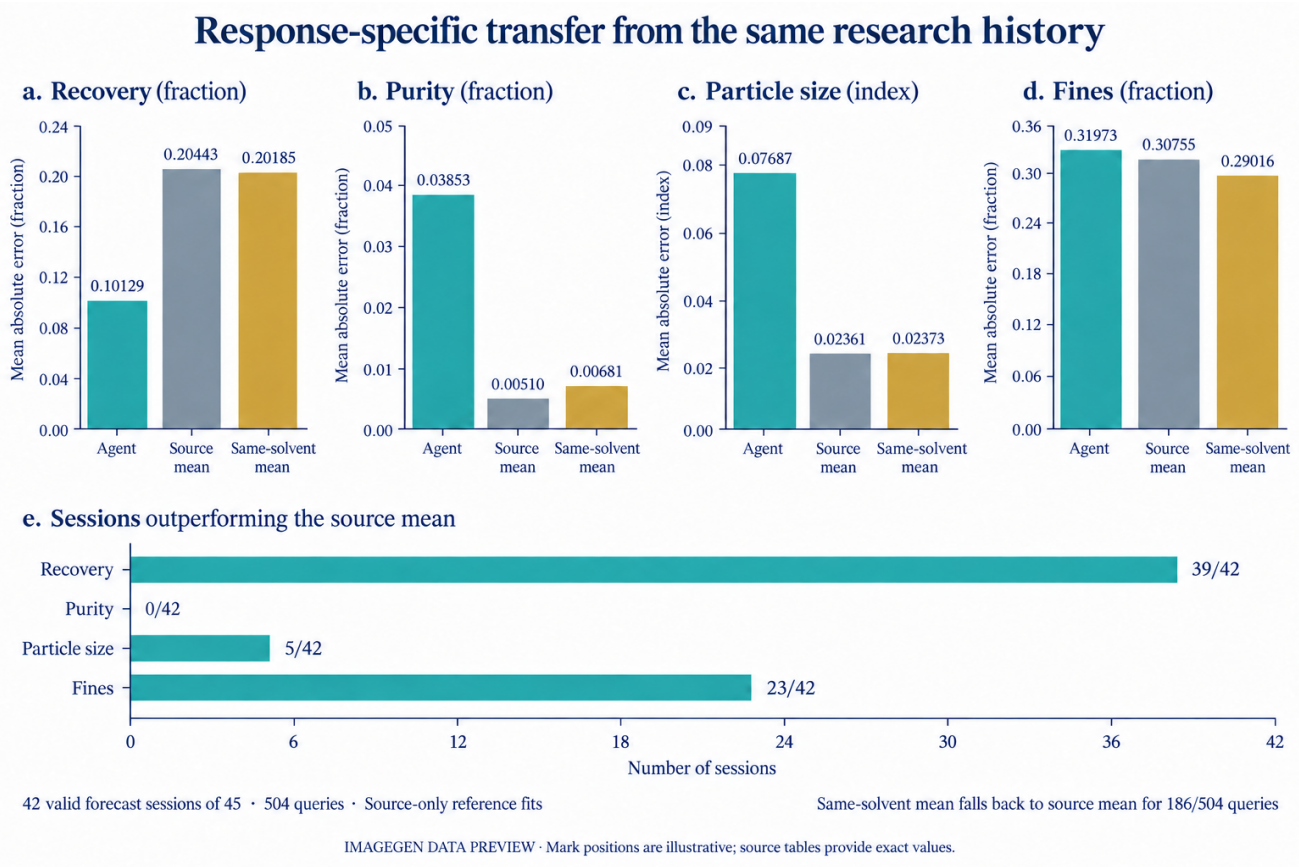
Source observations, versioned notes, sealed assessments, replay records and machine summaries are retained in the project evidence records. Supplementary Information reports current session outcomes, missingness, structured-case locations and retained broad-study tables. The integrated public archive and persistent identifier remain to be finalized; an earlier platform release should not be assumed to contain these later cohorts. Environment, execution and analysis code are maintained in the ChemWorld project. Manuscript compilation reads retained results without new model or simulator experiments.

Figures are temporary ImageGen previews: one framework illustration and six data-layout previews. Numerical anchors come from retained evidence; generated marks remain illustrative. Exact results are reported in the text and tables. Prompts and provenance accompany the artwork.

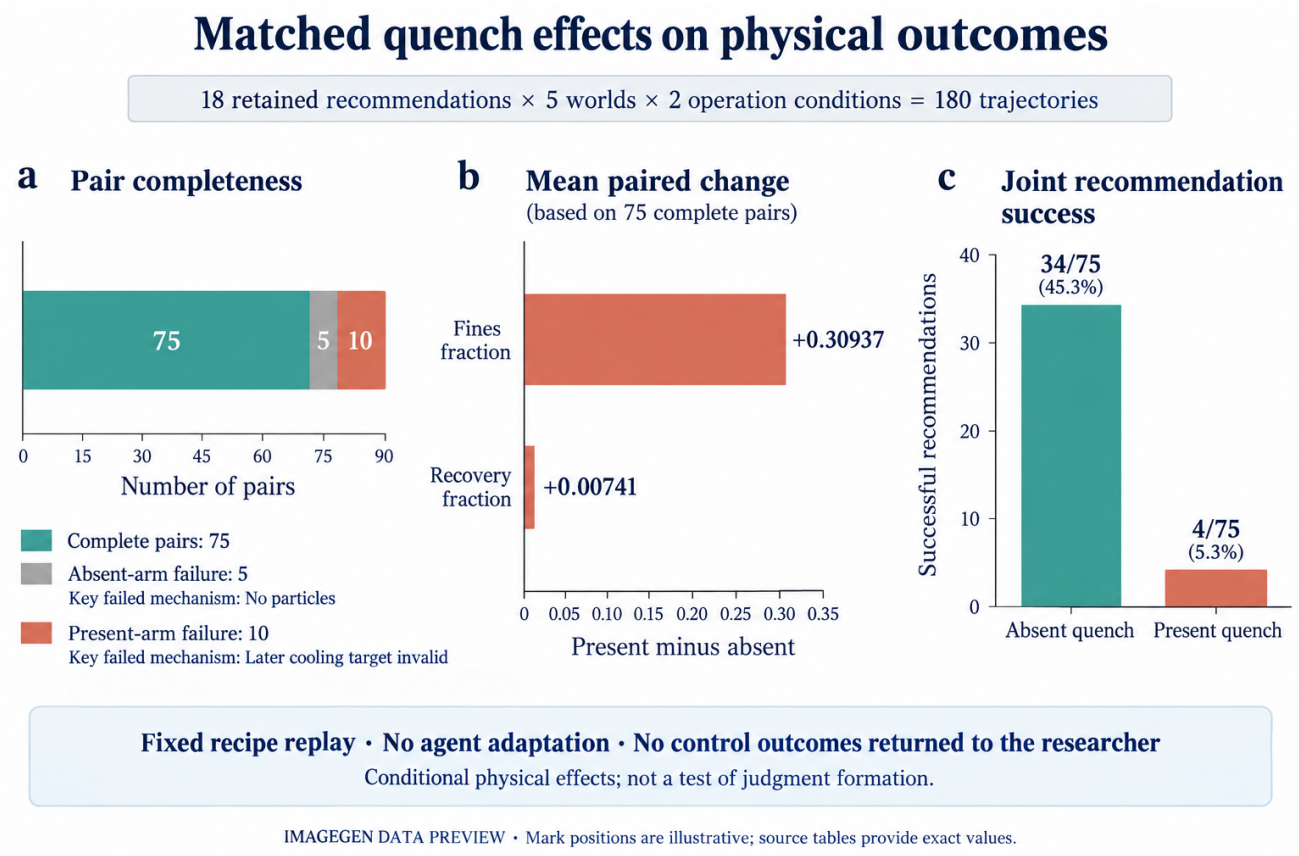
Extended Data



Extended Data Figure 1 | Coverage and repeatability of process evidence. ImageGen data preview. a, Current equilibrium/crystallization sessions number 45/45, complete chains 39/41 and valid forecasts 41/42. b, There are 34/58 process windows across 30/35 sessions; the remaining 15/10 sessions have no complete observable window, which does not establish absence of learning. c, Fifty-two corrected-crystallization immediate plans match subsequent execution and six partly match; this is not a learning-success rate. d, Across fifteen world-by-information cells with three sessions each, joint recommendation outcomes are six all-pass, one all-fail, five mixed and three incomplete. Earlier protocol-affected crystallization is not pooled. Generated bar positions are illustrative; exact denominators remain in the tables.



Extended Data Figure 2 | Response-specific transfer from the same research history. ImageGen data preview. a–d, Agent forecasts and two source-only references are compared for recovery, purity, particle-size index and fines, retaining each response's own public scale. Particle size is the dimensionless index min(d50/250 micrometres, 1). e, Agent forecasts outperform the overall source mean in 39/42, 0/42, 5/42 and 23/42 sessions, respectively. The 42 valid corrected-crystallization sessions supply 504 forecast queries. Same-solvent means fall back to the overall mean in 186/504 queries and do not match thermal history or quantities. Neither reference is fitted to held-out truth. Generated geometry is illustrative; five-world-equal numerical errors are reported in the supplementary tables.



Extended Data Figure 3 | Matched quench effects on physical outcomes. ImageGen data preview. Eighteen historical recommendations are replayed in five worlds with and without one quench operation, yielding ninety pairs and 180 trajectories. a, Seventy-five pairs yield both final assays; five absent arms lack particles and ten present arms invalidate a subsequent fixed cooling target. b, Complete-pair mean changes are +0.30937 in fines and +0.00741 in recovery. c, Joint success changes from 34/75 without quench to 4/75 with quench. These are conditional physical effects under the initial-charge recovery definition. The agent neither adapts nor receives control outcomes, so this intervention does not identify judgment formation. Generated geometry is illustrative and all failed pairs remain reported.